

Nicholas Eisenberg, Ph.D.

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Experience

Senior Scientist - Machine Learning

June 2022 - Present

Nevada National Security Site

- **LLM Inference Server (Piccata):** Architected, deployed, and administer a multi-user LLM platform on a RHEL GPU server, exposing internally hosted models through OpenAI-compatible vLLM APIs. Built a FastAPI gateway for authentication, routing, usage tracking, and persistence; developed `libserverctl` for user, API-key, certificate, and usage administration; and built a `Streamlit` portal for access requests and provisioning API tokens and client certificates. Deployed the platform with rootless Podman, Nginx reverse proxying and TLS/mTLS.
- **LLM-Assisted Image Annotation and Retrieval (Photo-Pilot):** Designed and deployed a `Streamlit` application that enables users to annotate and tag images with the assistance of multimodal LLMs hosted on Piccata, persist image annotations and metadata in a SQL database, and query the resulting image collection. Deployed the application on a RHEL server using Podman, with Nginx providing reverse proxying.
- **Python LLM Client Library (ChatLib):** Developed and maintain an internal Python library built on the OpenAI SDK that simplifies interaction with internally hosted LLMs, providing conversation management, streaming, multimodal text and image inputs, response formatting, and a consistent interface for scientific and engineering applications.
- **Internal Scientific Software Monorepo:** Maintain a large internal monorepo supporting ML and scientific applications, including shared Python libraries, dependency and release workflows, automated testing, CI/CD, and Linux deployment infrastructure.
- **Retrieval-Augmented Generation (RAG):** Designed and deployed a RAG system for internal Python/C++ codebases and technical documents, implementing language-aware ingestion, semantic embeddings, HNSW vector retrieval, and locally hosted LLM inference.

Technical Skills

Programming Languages: Python, Bash, Lua, SQL, Cypher, $\text{\LaTeX}$

Libraries: PyTorch, Pandas, FastAPI, SQLAlchemy, Streamlit, Matplotlib, vLLM

Tools / Infrastructure: Linux (RHEL, Ubuntu), Podman, Docker, Nginx, Git, CI/CD, SLURM, AWS

Open-Source Projects

iron.nvim - Sole Maintainer

May 2024 - Present

github.com/Vigemus/iron.nvim

Sole maintainer of a cross-platform Neovim plugin written in Lua for interactive REPL integration. Own the project lifecycle including feature development, code review, releases, regression testing, issue triage, and technical support across Linux, macOS, and Windows.

Education

Auburn University - Ph.D.

June 2022

Research Area: Stochastic Partial Differential Equations

Advisor: Le Chen

University of Nevada, Las Vegas - B.S./M.S.

December 2015 / January 2018

Applied Mathematics

Publications

- 1: Eisenberg, N., Chen, L. *Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics*. Stoch PDE: Anal Comp (2022).
- 2: Eisenberg, N., Chen, L. *Invariant Measures for the Nonlinear Stochastic Heat Equation with No Drift Term*. J Theor Probab (2024).